XRPL Validator — Daily Engineering Report · 2026-08-23

Period: 2026-08-23 00:00 to 24:00 US Eastern (2026-08-23 04:00 to 2026-08-24 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: xrpl-validator merge records (they begin 2026-07-04). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data; data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 0 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(xrpl-ledger): tecKILLED discards the OfferSequence cancel — fifty-three consecutive ledgers** (`808c980`, xrpl-validator) OfferCreate is not a standard tec-rollback transactor: doApply keeps TWO sandboxes and picks by applyGuts' second return — sb (everything) or sbCancel (fee + the unfunded/expired reaps flowCross mirrors into both, OfferCreate.cpp:463-465, 969-989). Every return says {_, true} EXCEPT the two tecKILLED and one tefINTERNAL — and the OfferSequence cancellation is written to sb ONLY (:647). A killed IoC/FoK therefore …
- **fix(xrpl-ledger): tfSell residual gets rounds half-even, not truncated — fifty-four consecutive ledgers** (`4cffac5`, xrpl-validator) rippled's afterCross.in for a tfSell remainder is an STAmount SUBTRACTION (takerGets minus the crossed in), and Number normalizes the result to 16 digits at to-nearest. Our residual took norm16 — a truncation.
- **fix(xrpl-ledger): payment remainders from savedOuts totals; want_gross via mulRatio; fee trim reuses the rev cache — sixty-seven consecutive ledgers** (`dae7e88`, xrpl-validator) Three coupled one-ulp rules, all invisible to key-level MATCH — the census and the replay byte-diff are the legs that see them: 1. TOTALS, NOT RUNNING REMAINDERS. StrandFlow keeps every winning pass in a sorted multiset and re-derives remainingOut = outReq − sum(savedOuts) and remainingIn = sendMax − sum(savedIns) every iteration (StrandFlow.h:639-642, 762-766; sum folds ascending, one half-even 16-digit add per …
- **feat(xrpl-ledger): limitOut quality-function trim, net-units judge, and per-iteration NET destination credits — seventy-two consecutive ledgers** (`677f310`, xrpl-validator) The tfLimitQuality campaign, closing scout-find 106453302 (the extra-reap class) and stretching the replay streak 67 → 72. Three mechanisms, one specimen: BFC61DEF (16 RLUSD-family arb, XRP→USDC→USDT→ETH, limit set).
- **fix(xrpl-ledger): a zero hop-carry delta with nonzero walk-accounting is a BLIND delta, not a dry hop — the self-fill face** (`d882679`, xrpl-validator) #106455105 1232F8E8 / #106455106 E71A9888 (one bot, RLUSD→BTC→XRP, DeliverMin, tfPartialPayment): mainnet fills 16 RLUSD → 0.000203 BTC → 10879958

drops, lands under DeliverMin 10913307, and rolls back the flow as tecPATH_PARTIAL (fee-only). We answered tecPATH_DRY.

- **fix(xrpl-ledger): gross_in is mulRatio and in-limited fills are GROSS-PRIMARY — the coupled spend-side campaign, seventy-five consecutive ledgers** (`a12ffd5`, xrpl-validator) The campaign the gross_in note parked ("one campaign, not piecemeal") — both halves receipt-pinned, and the A/B proved neither works alone: 1. gross_in: exact ceil → rippled mulRatio (BookStep.cpp:770 stpIn = mulRatio(ofrIn, trIn, QUALITY_ONE, roundUp) — nearest-16 + bump on a pre-normalize remainder). #106455107 7844199C: net 0.01536809504818859 × 1.003 = …315|577 → nearest …316 → bump …317 (mainnet's spend-line …
- **fix(xrpl-ledger): the hop-joint net division is mulRatio-NEAREST, not floor — seventy-six consecutive ledgers** (`e4c1581`, xrpl-validator) The morning's deferred F1 site, now specimen-pinned. rippled nets an in-limited carry through a gateway rate at mulRatio-nearest (DirectStepI fwd srcToDst = mulRatio(in, QUALITY_ONE, srcQOut, false); the LIMITSTEPIN receipts show the same division inside BookStep: stpIn = the gross cap verbatim, ofrIn = the net by division). Our hop joint floored.
- **fix(xrpl-ledger): merge_pages sorts by compare_tokens, not whole-ID hex — eighty-one consecutive ledgers** (`3b38011`, xrpl-validator) #106455111 D9C28F93 (NFTokenAcceptOffer): the transfer consolidates the buyer's page chain (two pages modified, one deleted, survivor at exactly 32 tokens). rippled's merge concatenates two adjacent pages — each already a compareTokens-ordered run, low 96 BITS first, whole ID only as the tiebreak — and never re-sorts. Our merge_pages re-sorted the union by the whole-NFTokenID hex string, reordering across the seam …
- **fix(xrpl-ledger): direct and run-tail asks take the NET twin verbatim — dividing the gross pot cannot recover it — eighty-four consecutive ledgers** (`b14b09a`, xrpl-validator) #106455116 F04EF64E (rapido5, XRP→RLUSD→EUR.rhub8 →run→ EUR.rvYAfW, DeliverMin): the mixed strand's tail run was asked for net_rem_out = want_gross / want_rate — and that division can NEVER recover the Amount: want_gross was rounded UP past Amount × rate (mulRatio, a12ffd5's own fix), so the quotient sits ABOVE the Amount (… 2710/1.0015 = 2588523229458522.2) and any rounding lands high. The run then planned and …
- **fix(xrpl-ledger): a book strand's sin is GROSS — the round loop drained the NET pot with it — one hundred seven consecutive ledgers** (`afb53c1`, xrpl-validator) #106455119 EFFA952F (default-path FLR→XRP, SendMax 4018 FLR at the 1.003 issuer, partial): the strand-level spent chooser (1f31546) deliberately trusts the balance DELTA on a fee-bearing spend leg — and the line loses net + fee, so a book strand's sin comes back GROSS (round 0: 1339.4425643283, rippled's stpIn verbatim). The dispatch still classified it in_gross=false, so the update block: - subtracted the GROSS …
- **fix(xrpl-ledger): every spend-side net division is mulRatio-NEAREST — the live-line clamp was the fourth and last floor — one hundred sixteen consecutive ledgers** (`737f067`, xrpl-validator) #106455142 C0442CB5 (default-path UNI→XRP drain, SendMax at the 1.001 issuer): the sender's whole 2.500399999999944 UNI line nets to quotient …041|958 — rippled's implied first DirectStep says …042 (nearest); our fund-limit clamp against the LIVE line balance still floored to …041, shaved the walk's budget one ulp under, and the maker offer 5622E947's residual (wants0 − net) read …959 for mainnet's …958. Both UNI …

Independence program

No independence record for UTC day 2026-08-23; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Multi-Asset Basket Vaults** (amm-defi) Add a tfBasket flag to VaultCreate letting a SingleAssetVault own an array of up to 8 sfVaultAsset entries (asset+issuer plus sfTargetWeight), each ba
- **Selective Disclosure for Confidential Transfers** (privacy) Add a ttDISCLOSE_COMMITMENT transaction carrying sfSubjectTxnID (a settled ConfidentialTransfer payment) and sfAuditorID, valid only when sfAuditorID
- **DID-Attested Reputation Credit Lines** (lending) Add sfRepaymentStreak (uint32) to the borrower position record inside a LendingProtocol Vault's loan object, incremented on each on-time LoanRepay and

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
08-23 04:36	—	xrpl-validator	`808c980`	fix(xrpl-ledger): tecKILLED discards the OfferSequence cancel — fifty-three consecutive ledgers
08-23 05:22	—	xrpl-validator	`4cffac5`	fix(xrpl-ledger): tfSell residual gets rounds half-even, not truncated — fifty-four consecutive ledgers
08-23 15:11	—	xrpl-validator	`dae7e88`	fix(xrpl-ledger): payment remainders from savedOuts totals; want_gross via mulRatio; fee trim reuses the rev cache — sixty-seven consecutive ledgers
08-23 16:46	—	xrpl-validator	`677f310`	feat(xrpl-ledger): limitOut quality-function trim, net-units judge, and per-iteration NET destination credits — seventy-two consecutive ledgers
08-23 17:53	—	xrpl-validator	`d882679`	fix(xrpl-ledger): a zero hop-carry delta with nonzero walk-accounting is a BLIND delta, not a dry hop — the self-fill face

Committed	Merged into main	Repo	Commit	Subject
08-23 19:08	—	xrpl-validator	`a12ffd5`	fix(xrpl-ledger): gross_in is mulRatio and in-limited fills are GROSS-PRIMARY — the coupled spend-side campaign, seventy-five consecutive ledgers
08-24 00:05	—	xrpl-validator	`e4c1581`	fix(xrpl-ledger): the hop-joint net division is mulRatio-NEAREST, not floor — seventy-six consecutive ledgers
08-24 00:59	—	xrpl-validator	`3b38011`	fix(xrpl-ledger): merge_pages sorts by compare_tokens, not whole-ID hex — eighty-one consecutive ledgers
08-24 01:42	—	xrpl-validator	`b14b09a`	fix(xrpl-ledger): direct and run-tail asks take the NET twin verbatim — dividing the gross pot cannot recover it — eighty-four consecutive ledgers
08-24 02:28	—	xrpl-validator	`afb53c1`	fix(xrpl-ledger): a book strand's sin is GROSS — the round loop drained the NET pot with it — one hundred seven consecutive ledgers
08-24 03:25	—	xrpl-validator	`737f067`	fix(xrpl-ledger): every spend-side net division is mulRatio-NEAREST — the live-line clamp was the fourth and last floor — one hundred sixteen consecutive ledgers

Times in UTC. A commit appears on the day it was committed to main or merged into it.